Equations

Formulas and Equations

2026

Preliminary Mathematics Standard

Outline

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What is a Linear Equation?

Definition
A **linear equation** has:

- Variables with exponent 1 only
- No products of variables
- Constant coefficients

Example

$$ax + b = 0$$

where $a \neq 0$

Examples

- $3x + 5 = 17$
- $2x - 7 = x + 3$
- $\frac{x}{4} + 2 = 10$
- $5(2x - 1) = 3x + 12$

Non-Examples

- $x^2 + 3x = 5$ (quadratic)
- $xy = 12$ (product of variables)
- $\sqrt{x} = 4$ (not linear)

Translating Words to Mathematics

Words	Math Symbol
is, equals, gives	$=$
sum, total, plus, increased by	$+$
difference, minus, decreased by	-
product, times, multiplied by	$\times$ or $\cdot$
quotient, divided by, per	$\div$ or fraction
more than	$+$
less than	-
twice, double	$2\times$
half, one-half	$\frac{1}{2}\times$

Example

- "5 more than x": $x + 5$
- "Twice a number": $2x$
- "3 less than y": $y - 3$
- "Half of n": $\frac{n}{2}$

Watch Out!

- "5 less than x" $= x - 5$ NOT $5 - x$
- Order matters!

5-Step Problem Solving Process

1. **Read & Understand**
    - What are you asked to find?
    - What information is given?
2. **Define Variables**
    - Let $x = \ldots$
    - Label clearly
3. **Write Equation**
    - Translate words to algebra
    - Use key words table
4. **Solve Equation**
    - Use algebraic methods
    - Show all working
5. **Interpret Solution**
    - Answer in context
    - Check reasonableness

Example 1: Simple Number Problem

Example
Three more than twice a number is 17. Find the number.

Step 1: Understand
We need to find an unknown number.

Step 2: Define Variable
Let $x =$ the number

Step 3: Write Equation

- "Twice a number": $2x$
- "Three more than": $+3$
- "is 17": $= 17$

$$2x + 3 = 17$$

Example 1: Solving & Interpreting

Step 4: Solve Equation

$$2x + 3 = 17$$
$$2x + 3 - 3 = 17 - 3$$
$$2x = 14$$
$$\frac{2x}{2} = \frac{14}{2}$$
$$x = 7$$

Step 5: Interpret Solution

- The number is 7
- Check: Twice 7 is 14, plus 3 is 17
- This is a reasonable solution

Key Insight
The equation $2x + 3 = 17$ represents the word problem exactly!

Example 2: Age Problem

Example

Jane is 5 years older than Tom. In 3 years, the sum of their ages will be 31. Find their current ages.

Define Variables

Let:

- $T =$ Tom's current age
- $J =$ Jane's current age

Given Information

- Now: $J = T + 5$
- In 3 years:
  - Tom: $T + 3$
  - Jane: $J + 3$
- Sum $= 31$

Example 2: Solving

Write Equation

$$(T + 3) + (J + 3) = 31$$

But $J = T + 5$, so:

$$(T + 3) + ((T + 5) + 3) = 31$$

Simplify:

$$(T + 3) + (T + 8) = 31$$

$$2T + 11 = 31$$

Solve

$$\begin{aligned} 2T + 11 &= 31 \\ 2T &= 20 \\ T &= 10 \end{aligned}$$

$$J = T + 5 = 10 + 5 = 15$$

Example 2: Interpretation & Check

Solution

- Tom is 10 years old
- Jane is 15 years old

Check

- Jane is 5 years older than Tom: $15 - 10 = 5$
- In 3 years:
  - Tom: $10 + 3 = 13$
  - Jane: $15 + 3 = 18$
  - Sum: $13 + 18 = 31$

Important

- Age problems often involve "now" and "future/past"
- Define ages carefully
- Check in both time frames

Example 3: Measurement Problem

Example

A rectangle has length 3 cm more than its width. If the perimeter is 26 cm, find the dimensions.

Step 1 & 2

- Need: width and length
- Let $w =$ width (cm)
- Length: $w + 3$ (cm)

Perimeter Formula

$$P = 2 \times \text{length} + 2 \times \text{width}$$

$$P = 2l + 2w$$

Step 3: Write Equation

$$2(w + 3) + 2w = 26$$

Example 3: Solving

Solve Equation

$$
\begin{aligned}
2(w+3)+2w &= 26 \\
2w+6+2w &= 26 \\
4w+6 &= 26 \\
4w &= 20 \\
w &= 5
\end{aligned}
$$

Find Length

$$l = w+3 = 5+3 = 8$$

Interpret

- Width $= 5$ cm
- Length $= 8$ cm
- Check: Perimeter $= 2(8)+2(5) = 16+10 = 26$
- Reasonable dimensions

Common Mistakes to Avoid

Mistake #1

- **What**: Wrong variable order
- **Example**: "5 less than x"
  - Correct: $x - 5$
  - Wrong: $5 - x$

Mistake #2

- **What**: Forgetting units
- **Fix**: Always include units in answer

Mistake #3

- **What**: No interpretation
- **Fix**: Always answer in context

Mistake #4

- **What**: Not checking reasonableness
- **Example**: Negative age, negative length
- **Fix**: Ask "Does this make sense?"

Your Turn: Practice Problem

Example

A taxi charges \$3.50 flag fall plus \$2.10 per kilometre. If the fare is \$25.70, how far did the taxi travel?

Think About:

1. What are you asked to find?
2. What information is given?
3. What variable should you use?
4. What equation represents this situation?

$$\text{The total cost} = \text{fixed cost} + (\text{rate} \times \text{distance})$$

Practice Problem Solution

Step 1 & 2

- Find: distance travelled (km)
- Let d = distance (km)

Step 3: Equation

$$3.50 + 2.10d = 25.70$$

Step 4: Solve

$$\begin{aligned} 2.10d &= 25.70 - 3.50 \\ 2.10d &= 22.20 \\ d &= \frac{22.20}{2.10} \\ d &= 10.57\ldots \approx 10.6 \text{ km} \end{aligned}$$

Example

Challenge Question "The sum of three consecutive even numbers is 78. Find the numbers."

Let x be middle number

$$x-2+x+x+2=78$$

$$3x=78$$

$$x=26$$

24, 26, 28

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Homework

Complete all questions from *Formulas and Equations — Equations*.